



GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING

COURSE CODE: CM5467

COURSE TITLE: COMPUTER SECURITY

TEACHING SCHEME:

LEVEL OF COURSE	PRERE-QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	CM3404	03	-	-	03	16	48	-	-

EXAMINATION SCHEME:

ESE PAPER HRS.	THEORY(Marks)				PRACTICAL(Marks)		TOTAL (Marks)
	ESE		PA	TOTAL	ESE	PA	
	MAX.	70	30*	100	-	-	100
3Hrs	MIN.	28	---	40	-	-	40

@: Internal Assessment #: External assessment-Practical based \$: online examination

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1. RATIONALE:

Computer security, one of the most important and relevant area of computing today. Now a days due to various threats designing security in organization is an important consideration. It is essential to understand basic security principles, various threats to security and techniques to address these threats. The student will be able to recognize potential threats to confidentiality, integrity and availability and also able to implement various computer security policies.

This course will introduce basic cryptographic techniques, fundamentals of computer/network security, Risks faced by computers and networks, security mechanisms, operating system security, secure System design principles, and network security principles. It will develop knowledge for security of information within organizations.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

- 1) Identify risks faced by Computer Systems and Information hazard in various situations.
- 2) Identify role of people in computer security.
- 3) Apply cryptographic algorithms and protocols to maintain Computer Security.
- 4) Suggest Type of firewall & intrusion detection system for computer network.
- 5) Authenticate users using security protocols.
- 6) Identify different data recovery mechanisms and cyber-crimes.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hrs	Marks
Unit 1. Introduction to computer security	1a. Explain the importance of given pillars of computer security. 1b. Explain the characteristics of given type of threat. 1c. Explain types of attacks related with security. 1d. Identify different techniques to secure OS and applications	1.1 Foundations of Computer Security –Definition and Need of computer security, Security basics: Confidentiality, Integrity, Availability, Accountability, Non-repudiation, Reliability. 1.2 Risk and Threat Analysis: Assets, Vulnerability, Threats, Risks, Counter measures. 1.3 Threat to Security: Viruses, Phases of Viruses, Types of Virus, Dealing with Viruses, Worms, Trojan Horse, Intruders, Insiders. 1.4 Type of attacks: Active and Passive attacks, Denial of service, DDOS, backdoors and trapdoors, sniffing, spoofing, man in the middle, replay, TCP/IP Hacking, encryption attacks. 1.5 Operating system security: Operating system hardening, general steps for securing operating system, updates: hotfix, patch, service pack.	1	06	08
Unit 2. User Authentication & Access Control	2a. Explain given method to guess the passwords. 2b. Explain the given method of Biometric. 2c. Explain Authentication and Authorization with example. 2d. Describe the features of given access control policy.	2.1 Identification and Authentication: Username & Password, Guessing password, Password attacks- Piggybacking, Shoulder surfing, Dumpster diving. 2.2 Biometrics: finger prints, hand prints, Retina, patterns, voice patterns, signature and writing patterns, keystrokes. 2.3 Access controls: Definition, Authentication Mechanism, principle-Authentication, Authorization, Audit, Policies: DAC, MAC, RBAC.	2	08	12

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hrs	Marks
Unit 3 Cryptography	3a. Define terms related to cryptography. 3b. Encrypt/Decrypt the given text using different substitution techniques. 3c. Convert plain text to cipher text and Vice versa using various transposition techniques. 3d. Explain different techniques of cryptography.	3.1 Introduction: Plain Text and Cipher Text, Cryptography, Cryptanalysis, Cryptology, Encryption, Decryption. 3.2 Substitution techniques: Caesar's cipher, monoalphabetic and Poly-alphabetic 3.3 Transposition techniques: Rail fence technique, simple columnar, Steganography, Hashing. 3.4 Symmetric and Asymmetric cryptography: Introduction to Symmetric encryption, DES (Data encryption Standard) algorithm, Asymmetric key cryptography: Digital Signature.	3	10	12
Unit 4. Firewall and Intrusion Detection System	4a. Compare different types of firewall. 4c. Enlist configuration steps of firewalls. 4d. Describe VPN architecture. 4e. Understand intrusion and intrusion detection system.	4.1 Firewalls: Need for Firewall, limitations, characteristics, Types of Firewall :Hardware, Software, Packet filter, Proxy Server, Hybrid, Application gateways, circuit level gateway, Implementing Firewall. 4.2 Firewall Policies, configuration, limitations, DMZ. 4.3 Virtual Private Network- Introduction, architecture. 4.4 Intrusion Detection System- Intrusion detection systems (IDS), host based IDS, Network based IDS, Protocol based IDS, Signature based detection, Anomaly – based detection, Honey pots.	4	08	12
Unit 5. Network Security	5a. Explain the given authentication protocols. 5b. Explain different ways to implement IP Security 5c. Explain protocols related to Email security. 5d. Describe Public Key	5.1 Kerberos. 5.2 IP Security- overview, Protocols- AH, ESP, Modes- transport & Tunnel. 5.3 Email security- Email security standards, working principle of SMTP, PEM, PGP, S/MIME 5.4 Public key infrastructures: basics, digital certificates, certificate authorities,	5	10	14

	Infrastructure.	registration authorities, Trust models (Hierarchical, peer to peer, hybrid)			
Unit 6. Data Recovery Techniques & Cyber Crime	6a. Understand basics of recovery and cyber crimes. 6b. Implement different recovery techniques to recover sensitive information. 6c. Understand about different types of cyber crimes and law to prevent such crimes	6.1 Recovery: Introduction to Deleted File Recovery, Formatted Partition Recovery, Data Recovery Tools, Data Recovery Procedures and Ethics. 6.2 Cyber Crimes: Introduction, Hacking, Types of Hacking, Cracking, Pornography, Software Piracy, Intellectual property, Legal System of Information Technology, Mail Bombs, Bug Exploits, Cyber Crime Investigation, Ethical Hacking. 6.3 Cyber Laws: Introduction to IT act 2000 and IT act 2008s.	6	06	12

4. SUGGESTED STUDENT ACTIVITIES

Following are the suggested student related *co- curricular* activities which can be undertaken to accelerate the attainment of the various outcomes in this course:

- Prepare a list of classes (API) available in java and .NET for implementing cryptography and security
- Apply different security policies and strategies to secure college computers and network
- Apply and configure application hardening techniques to secure OS, Applications and websites
- Undertake micro-projects.

5 . SUGGESTED INSTRUCTIONAL STRATEGIES

These are sample strategies, which the teacher can use to accelerate the attainment of the various learning outcomes in this course:

- Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- Guide student(s) in undertaking micro-projects.
- Encourage students to refer different websites to have deeper understanding of the subject.
- Arrange visit to cyber crime departments of different organizations
- Arrange workshop on cyber crimes and ethical hacking
- Arrange expert lecture or seminar on latest trends in computer security.
- 'About **10-15% of the topics/sub-topics** which is relatively simpler or descriptive in nature is to be given to the students for **self-directed learning** and assess the development of the COs through classroom presentations (see implementation guideline for details).

6. SUGGESTED MICRO-PROJECTS.

Only one micro-project is planned to be undertaken by a student that needs to be assigned to him/her in the beginning of the semester. In all semesters, the micro-project are group-based (5-6 students) to build up skill and confidence in every student to become problem solver so that he/she contributes to the projects of the industry. The micro-project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each micro-project should encompass two or more COs which are in fact, an integration of UOs and ADOs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be

less than **16 (sixteen) student engagement hours** during the course. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs.

A suggestive list of micro-projects is given here. Similar micro-projects could be added by the concerned faculty:

- a. Case study on Cyber Security in India
- b. Big Data Technologies for Monitoring of Computer Security: A Case Study
- c. Case Studies in Secure Computing: Achievements and Trends
- d. Implement Client/Server communication using cryptography tools in your laboratory.
- e. Create digital certificate for your departmental/ personal communication.
- f. Implement communication system using steganography. Encrypt image and message using any cryptography technique.
- g. Three Level Password Authentication System
- h. Any other micro-projects suggested by subject faculty on similar line.

7. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks		
			R Level	U Level	A Level
1	Introduction to computer security	08	04	04	-
2	User Authentication & Access Control	12	06	06	-
3	Cryptography	12	-	04	08
4	Firewall and Intrusion Detection System	12	-	04	08
5	Network Security	14	06	04	04
6	Data Recovery Techniques & Cyber Crime	12	02	06	04
Total		70	18	28	24

R= Remember, U= Understanding, A=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

8. SUGGESTED LEARNING RESOURCES:

Sr. No.	Title Of Book	Author	Publication
1.	Computer Security	Dieter Gollmann	Third Edition ,Wiley Publication, India ISBN 13:978-8126550821
2.	Cryptography and Network Security	Atul Kahate	Third Edition. McGraw Hill Education, New Delhi ISBN 13:978-1259029882
3.	Computer Security Principles and Practices	William Stallings, Lawrie Brown	Fourth Edition, Pearson Education, India ISBN 13:978-0134794396
4.	Introduction to Computer Security	Matt Bishop	Second Edition ,Addison-Wesley, India ISBN 13:978-0321712332

9. SUGGESTED SOFTWARE/LEARNING WEBSITES

- a <http://nptel.ac.in/courses/106105162/>
- b. https://www.tutorialspoint.com//computer_security/computer_security_quick_guide.htm
- c. <http://learnthat.com/introduction-to-network-security/>
- d. <https://freevideolectures.com/course/3027/cryptography-and-network-security>
- e. <http://www.pgpi.org/doc/pgpintro>
- f. <http://www.emailtrackerpro.com>
- g. <http://www.kmint21.com>
- h. <http://www.jjtc.com/Steganography/tools.html>

10. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Mr.P.U.Malve	Lecturer in Computer Engineering.	Government Polytechnic Amravati
2	Miss.S.S.Chavhan	Lecturer in Computer Engineering.	Government Polytechnic Amravati

Government Polytechnic, Amravati Programme Board of Studies Computer Engineering has approved the above course curriculum on 14/05/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 14/07/2020

The Governing Body has approved the above course curriculum on 13/08/2021